Position z = -0.5 mm: same-position FRF validation

Experimental/model phases unwrapped independently; only one constant 360-deg display offset is allowed. No smoothing.
The validation split belongs to the same experimental record; nonconverged fits and anomalous records remain explicitly flagged.

Validation FRF z = -0.5 mm

$u_1 \to y_1$ $u_2 \to y_1$ $u_3 \to y_1$ $u_4 \to y_1$

$u_1 \to y_2$ $u_2 \to y_2$ $u_3 \to y_2$ $u_4 \to y_2$

$u_1 \to y_3$ $u_2 \to y_3$ $u_3 \to y_3$ $u_4 \to y_3$

$u_1 \to y_4$ $u_2 \to y_4$ $u_3 \to y_4$ $u_4 \to y_4$

Unwrapped phase (deg)

Frequency (Hz)